Bazil-uddin-khan OSCILLATION

$\tan\theta = \frac{v(0)}{x(0)}$

1). Horizontal spring:-

Q). A horizontal spring ($k = 300 N/m$) with mass $0.75kg$ attached to it undergoing simple harmonic calculate.

(a) Period $\rightarrow T = 2\pi\sqrt{m/k} \rightarrow 2\pi\sqrt{0.75/300} \rightarrow 0.3142s$

(b) frequency $\rightarrow \frac{1}{2\pi}\sqrt{k/m} \rightarrow \frac{1}{2\pi}\sqrt{\frac{300}{0.75}} \rightarrow 3.183Hz$

(c) angular frequency of oscillator $\rightarrow \omega = 2\pi f \rightarrow 2\pi \times (3.183) \rightarrow \boxed{20 rad/s}$

Q). Force of 500N is used to stretch a spring with 0.5kg mass attached to its by 0.35m.

(a) Value of k? $\rightarrow F = kx \rightarrow k = 500/0.35 \rightarrow \boxed{1428.6 N/m}$

(b) frequency? $\rightarrow \frac{1}{2\pi}\sqrt{\frac{1428.6}{0.5}} = 8.5Hz$

$\frac{f_2}{f_1} = \frac{\sqrt{k_2}}{\sqrt{k_1}}$

Q). 0.75kg mass vibrates. $x = 0.65\cos(7.35t)$. Determine

$x = A\cos(\omega t)$

$-0.65 \leq x \leq 0.65$ (a) amplitude $\rightarrow 0.65$

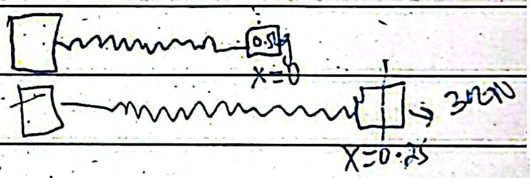
(b) frequency $\rightarrow \omega = 7.35 \rightarrow 7.35 = 2\pi f \rightarrow \boxed{f = 1.17Hz}$

(c) Period $\rightarrow T = 1/1.17 \rightarrow \boxed{0.855s}$

(d). spring constant $\rightarrow f = \frac{1}{2\pi}\sqrt{k/m} \rightarrow (2\pi f)^2 = k/m \rightarrow \boxed{k = (2\pi f)^2 \times m}$

2). ENERGY IN A SIMPLE Harmonic motion [max velocity and acceleration]:-

(1). A force of 300N is used to stretch a horizontal spring with 0.5kg attached to it by 0.25m. The block is released from rest and undergoes simple harmonic motion.



(a) spring constant $\rightarrow F = kx \rightarrow \boxed{k = 1200 N/m}$

(b). amplitude $\rightarrow$ max value of $x \rightarrow$ ~~A=~~ $A = 0.25$ $\quad -0.25 \leq x \leq 0.25$

(c) max acceleration $\rightarrow a_{max} = \frac{kA}{m} = \frac{1200(0.25)}{0.5} = 600 m/s^2$ or $F = ma$

(d). Mechanical energy $\rightarrow ME = \frac{1}{2}kA^2 \rightarrow \frac{1}{2} \times 1200 \times (0.25)^2 \rightarrow \boxed{37.5J}$

(e) Max velocity $\rightarrow ME = K.E + P.E \rightarrow \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \rightarrow 1200(0.25)^2 = 0.5v^2 \rightarrow \boxed{v = 12.25}$

$\boxed{(f)}$ velocity when $x = 0.15m \rightarrow V(x) = V_{max}\sqrt{1 - x^2/A^2} \rightarrow \sqrt{1 - \frac{(0.15)^2}{(0.25)^2}} = 9.8 m/s$

velocity at any position

or normal $M.E = K.E + P.E$

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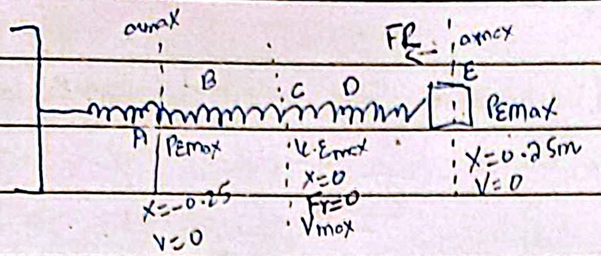
CamScanner



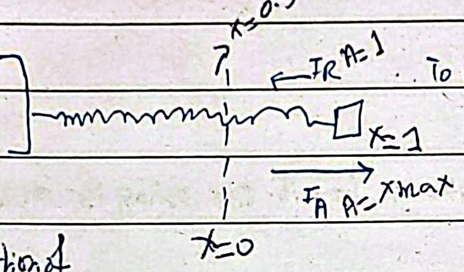
At Point A, E → acceleration is max. and M.E = P.E.

At C → M.E = K.E

At Point D, B → M.E = K.E + P.E

Simple harmonic motion:-

Mass spring system:-

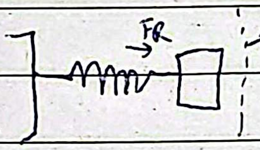


To move back and forth simple harmonic motion.

$F_R = -kx$ → restoring

x → "distance b/w equilibrium position and where it is placed"

"negative sign because FR is opposite to direction of motion"



→ To bring to mean position where $F_R = 0$

$F \propto x$ → $\boxed{\frac{F_1}{F_2} = \frac{x_1}{x_2}}$

[Note:- To find work required to stretch a spring] → $\boxed{\frac{kx^2}{2}}$

$k = 100\,N/m$

[friction opposes direction]

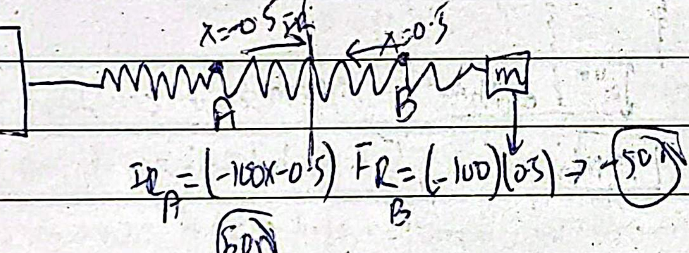
In 1 cycle → distance → 4x.

$v_{max} = \sqrt{\frac{k}{m}} \cdot A \cdot \sqrt{1 - x^2/A^2}$ → $V_{max} = \sqrt{\frac{k}{m}} \sqrt{1 - \frac{x^2}{A^2}}$

$a = \frac{k}{m}(x)$

$a = -\omega^2 x$

$a_{max} = \frac{k}{m}(A)$



$F_{R_A} = (-100 \times -0.5)$ $F_{R_B} = (-100)(0.5) \rightarrow -50N$

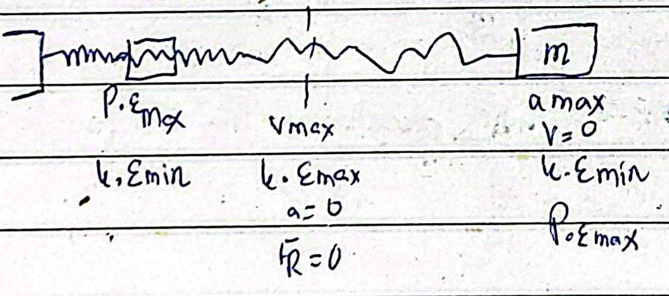
50N

$K.E = \frac{1}{2}mv^2$, $P.E = \frac{1}{2}kx^2$,

$M.E = K.E + P.E$

$\frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$

if $x = A$ → P.E is same as K.E → K



$x = 0$ → At equilibrium position

$\frac{1}{2}kA^2 = \frac{1}{2}mv^2$ → $V = \sqrt{\frac{k}{m}} \cdot A$

Frequency → $\frac{\text{No of cycles}}{\text{Time}}$

$T = \frac{\text{Time}}{\text{No of cycles}}$

Q). what happens to following if the maximum displacement of spring is doubled?

(a) Total energy of system → $\frac{1}{2} kA^2$ → Donot consider this → $A^2 \to (2)^2 \to$ [4 times]

(b). maximum velocity → $V_{max} = \sqrt{\frac{k}{m}} A' \to A' = 2$

(c). maximum acceleration → so → V_{max} is doubled.

→ $a = -\omega^2 x$ → so doubled.

v(x) point

→ $v(x) = \sqrt{1 - \frac{x^2}{a^2}}$

$F_R = -kx$

Donot neglect minus sign

Q). A spring streches 0.40m when a 2kg mass is hung from it. The spring is streched an additional 0.20 m from the equilibrium point and released.

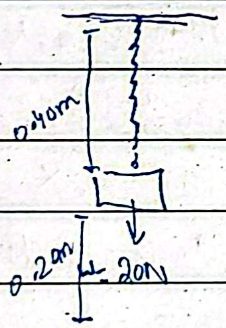
Determine (a) spring constant

(b) amplitude

(c) Max acceleration

(d). max velocity

(e) K.E, P.E, and ME when mass is 0.10 from it's equilibrium position?

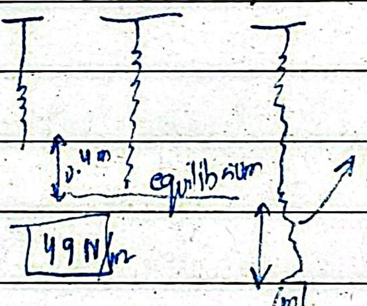


$m = 2kg$,

$F_x = mg$

$-kx = mg$

$k = \frac{(2)(9.81)}{0.4} = \boxed{49 N/m}$



$A = 0.20m$

As streched, released from this position

$a = 49(0.2) \to \boxed{4.9 m/s^2}$ $V_{max} = \sqrt{\frac{k}{m}} \times A \to \sqrt{49/2}(0.2) \to 0.99$

$P.E = \frac{1}{2} kx^2 \to \frac{1}{2}(49) \times (0.1)^2$

$ME \to \frac{1}{2} kA^2 \to A = 0.2$

$K.E = \frac{1}{2} mv^2$ $V(x) = V_{max} \sqrt{1 - x^2/A^2}$

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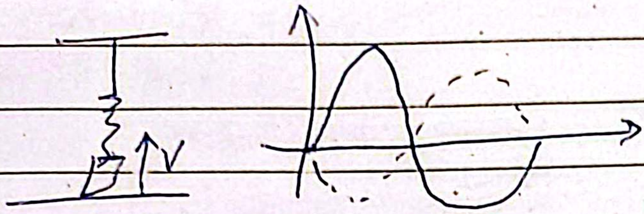
Q). A spring of force constant 300N/m vibrates with an amplitude of 45cm. When a 0.35kg mass hangs from it. What is eq of describing this motion?

(a) mass passes through equilibrium point with posi. velocity at $t=0$?



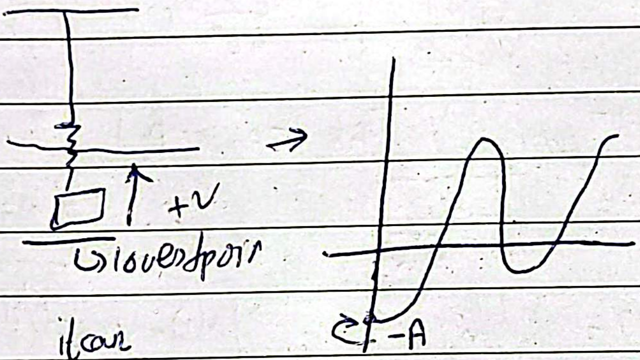
As it started from rest 0.

Moving up.

$x(t) = A\sin(2\pi ft)$.

if negative then $\Rightarrow x(t) = -A\sin(2\pi ft)$.

(b). mass starts at lowest point below equilibrium point with +ve velocity?



$x = -A\cos(2\pi ft)$

If function start from rest then if it goes up $\Rightarrow$ sin + else sin −

if it starts from top or bottom then it is $\rightarrow$ cos + else cos −

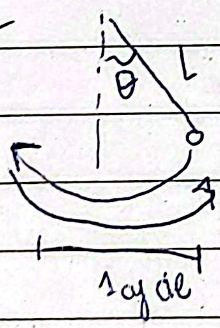
Due to internal friction damping is caused

if $f_A = f_N$ A ↑

f_R

$g = \frac{4\pi^2 L}{T^2}$

Simple Pendulum:



T = Time take complete 1 cycle $\rightarrow$ F = $\frac{\text{No of cycle}}{\text{time}}$

$T = 2\pi\sqrt{\frac{L}{g}}$

on moon $\rightarrow g = 1.6$

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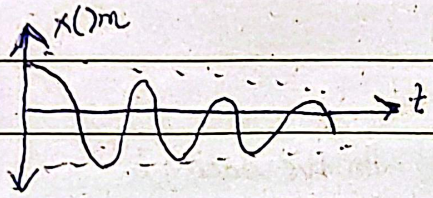
Damping:-

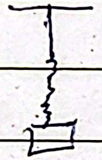
Decrease in amplitude with respect to time due to air resistance, internal friction.



Damping force $= -bv$ ($b \to$ constant, $v \to$ velocity)

$\Sigma F = ma \rightarrow -kx - bv = ma$

$ma + kx + bv = 0$



$[-kx] \rightarrow$ Restoring force
f_R

eq of motion $\left[m\frac{d^2x}{dt^2} + kx + b\frac{dx}{dt} = 0 \right.$

solution:-

$x(t) = Ae^{-\gamma t}\cos(\omega' t)$ [At $t=0$, $x=A$]

$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$ $\omega' \neq \omega = \sqrt{k/m}$

$\gamma = b/2m$ $\rightarrow \boxed{x(t) = Ae^{-b/2m\, t}\cos(\omega' t)}$

position function for a lightly damped

frequency:

$\omega' = 2\pi f'$

$f' = \frac{1}{2\pi}\sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$ $b^2 = 4mk$

overdamped $\rightarrow b^2 > 4mk$ critically damped $b^2 = 4mk$

underdamped $b^2 < 4mk$

$\rightarrow$ acceleration, and displacement opposite

$\rightarrow$ disp, ve, acceler $\rightarrow$ not in phase